Day 23: Binomial logistic regression

Data: moths and natural selection

Researchers glued equal numbers of light and dark morph moths to tree trunks at 7 locations from 0 to 51.2 km from Liverpool, where the trees were blackened by pollution. After 24 hours, they recorded how many moths had been removed (presumably eaten by predators).

- `morph`: light or dark
- `distance`: kilometers from Liverpool
- `Placed`: number of moths of that morph placed at that location
- `Removed`: number of those moths removed after 24 hours

morph	distance	placed	removed	proportion removed
light	0.0	56	17	0.304
dark	0.0	56	14	0.250
light	7.2	80	28	0.350
dark	7.2	80	20	0.250
light	24.1	52	18	0.346
dark	24.1	52	22	0.423
light	30.2	60	9	0.150
dark	30.2	60	16	0.267
light	36.4	60	16	0.267
dark	36.4	60	23	0.383
light	41.5	84	20	0.238
dark	41.5	84	40	0.476
light	51.2	92	24	0.261
dark	51.2	92	39	0.424

On HW8, we used the model:

```
model1 <- lm(log_odds_removed ~ distance + morph + distance * morph,
             data = moths)
```

What is the Y variable? What is the fitted equation?

Why doesn't this `glm` code work?

```
model1 <- glm(Removed ~ distance + morph + distance * morph,
              data = moths, family = "binomial")
```

```
Error:
! y values must be 0 <= y <= 1
```

The binomial response

Let Y_i be the _____ of successes (here: moths removed) out of n_i trials (here: moths placed) in row i. We assume the n_i trials in each row are independent and have the same probability of success π_i.

> **Binomial distribution**
>
> The sum of n independent success/failure trials, each with success probability π, follows a **Binomial**(n, π) distribution:
>
> $$E(Y) = _____ \qquad Var(Y) = _____$$
>
> A binary response is the special case $n =$ _____.

The binomial logistic regression model

The model is the same as before: it describes how the **probability** of success depends on the predictors, using the logit link.

$$\pi_i = P(Y = 1 | x_1, ..., x_p)$$

Log-odds scale:

$$\text{logit}(\pi_i) = \log\left(\frac{\pi_i}{1-\pi_i}\right) = _____ = \eta_i$$

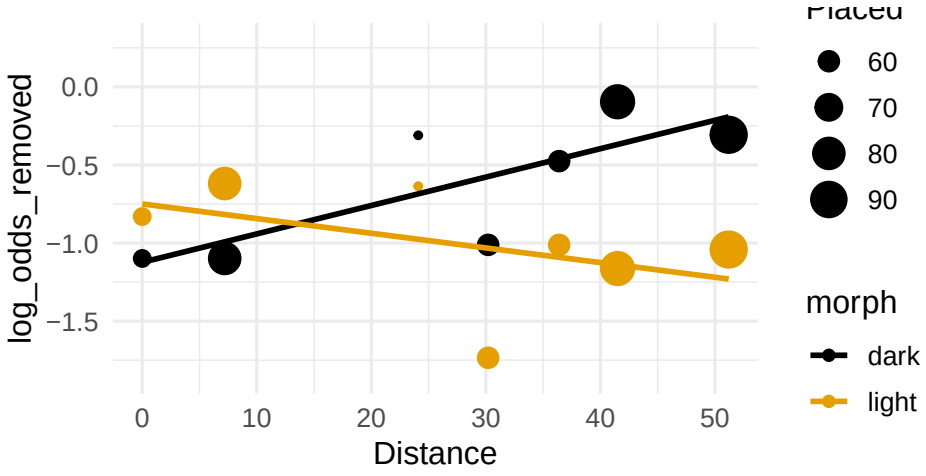
Probability scale:

$$____ = \frac{e^{\eta_i}}{1+e^{\eta_i}} = \frac{e^{\beta_0+\beta_1 x_{1i}+\beta_2 x_{2i}+\cdots+\beta_p x_{pi}}}{1+e^{\beta_0+\beta_1 x_{1i}+\beta_2 x_{2i}+\cdots+\beta_p x_{pi}}}$$

Response distribution: $Y_i \sim \text{Binomial}(n_i, \pi_i)$.

On the homework, we modeled the empirical log odds with `lm()`:

$$\widehat{\text{log_odds_removed}} = -1.123 + 0.018(\text{distance}) + 0.374(\text{morph}_{\text{light}}) - 0.028(\text{distance} \times \text{morph}_{\text{light}})$$



This approach required calculating the empirical log-odds "by hand" and adding to the dataset. Alternatively, we can fit with `glm()` using the `weights` argument

```
model3 <- glm(Removed/Placed ~ distance + morph + distance*morph,
              weights = Placed,
              data = moths,
              family = "binomial")
```

```
# A tibble: 4 x 5
  term                estimate std.error statistic      p.value
  <chr>                  <dbl>     <dbl>     <dbl>        <dbl>
1 (Intercept)          -1.13     0.198       -5.70 0.0000000117
2 distance              0.0185   0.00565      3.28 0.00105
3 morphlight            0.411    0.274        1.50 0.134
4 distance:morphlight  -0.0278   0.00809     -3.44 0.000588
```

Note: the estimates are slightly different due to different _____

Checking conditions

> **Binomial logistic regression conditions**
>
> - **Binomial counts**: Y_i is the number of successes in n_i trials
> - **Independence**: the trials (and the rows) are _____
> - **Linearity**: the _____ is a linear function of the predictors
> - **Variance structure**: $Var(Y_i) =$ _____, which is not constant, but is determined by the mean. We don't need an "equal variance" condition.

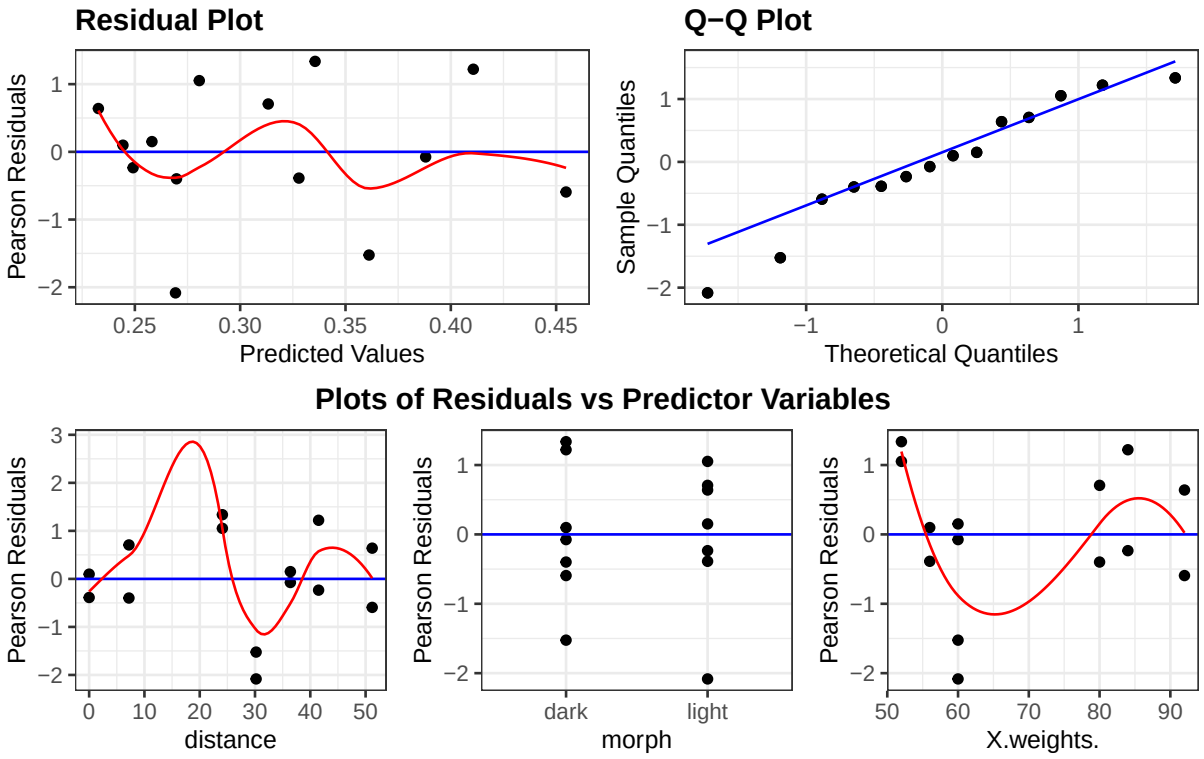
We check **linearity** with empirical logit plots (as in Day 20), but now every row is already a group, so we don't need to bin.

And we check linearity after accounting for the other predictors with **Pearson residuals**:

Pearson Residuals

$$\text{Pres}_i = \frac{Y_i - n_i\hat{\pi}_i}{\sqrt{n_i\hat{\pi}_i(1-\hat{\pi}_i)}}$$

For large enough n_i, Pearson residuals behave approximately like draws from a _____ distribution (unlike binary logistic regression, where the residuals were in two bands). We plot them against the fitted values and the predictors, with a smoother.



Because each row is a group, we can ask a question we couldn't ask before: **does the model fit the group proportions?**

Pearson χ^2

The **Pearson χ^2 statistic** compares the observed counts Y_i to the counts the model predicts, $n_i\hat{\pi}_i$:

$$X^2 = \sum_i \frac{(Y_i - n_i\hat{\pi}_i)^2}{n_i\hat{\pi}_i(1-\hat{\pi}_i)} = \sum_i \text{Pres}_i^2$$

- X^2 is _____ when the predicted counts are close to the observed counts (good), and _____ when they are far away (bad).
- If the model is correct and all the counts are large enough (about 5 or more successes *and*

failures in every row), $X^2 \dot{\sim} \chi^2$ with $df = n - (p + 1)$, where n is the number of **rows** (binomial observations), not the number of moths.

Goodness-of-fit test

Hypotheses: H_0: _____ (the model is adequate) vs. H_a: the model is inadequate
Test statistic: X^2 = the sum of the squared Pearson residuals
Reference distribution: χ^2 with $df =$ _____
p-value: the area _____ X^2 (upper tail). A *small* p-value is evidence of lack of fit.

```
X2 <- sum(residuals(model3, type = "pearson")^2)
X2
```

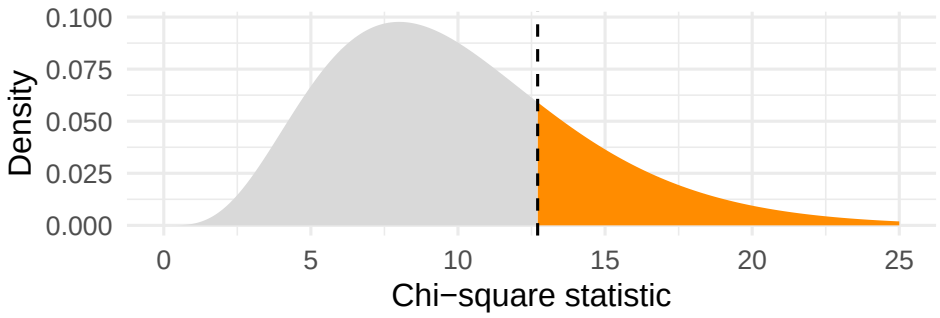
```
[1] 12.70859
```

```
df.residual(model3)
```

```
[1] 10
```

To find the p-value:

```
1 - pchisq(____, ____)
```



Warning

This goodness-of-fit test is only valid for **binomial** logistic regression with large enough n_i. It is *not* valid for binary logistic regression (each row has $n_i = 1$).

Summary

	Binary logistic	Binomial logistic
Response		
`glm()` response		
Residual plots		
Goodness-of-fit test valid?		

- Binomial logistic regression models the _____ of success for each row, using the same logit link as binary logistic regression.
- The variance of Y_i is _____, so it depends on π_i.
- **Pearson residuals** behave like $N(0, 1)$ draws when the n_i are large, so we can plot them against fitted values and predictors.
- The **Pearson X^2 statistic** is compared to a χ^2 distribution with $n - (p + 1)$ df to test **goodness of fit**.

R function reference

Function	Purpose
`glm(y/n ~ x, weights = n, data, family = "binomial")`	Fit a binomial logistic regression (proportion response with `weights`)
`glm(cbind(y, n - y) ~ x, data, family = "binomial")`	Same model, with a (successes, failures) response
`resid_panel(model, type = "pearson", smoother = TRUE)`	Pearson residual plots
`resid_xpanel(model, type = "pearson", smoother = TRUE)`	Pearson residuals against each predictor
`sum(residuals(model, type = "pearson")^2)`	Pearson X^2 statistic for the goodness-of-fit test
`1 - pchisq(X2, df)`	Upper-tail χ^2 probability (p-value)